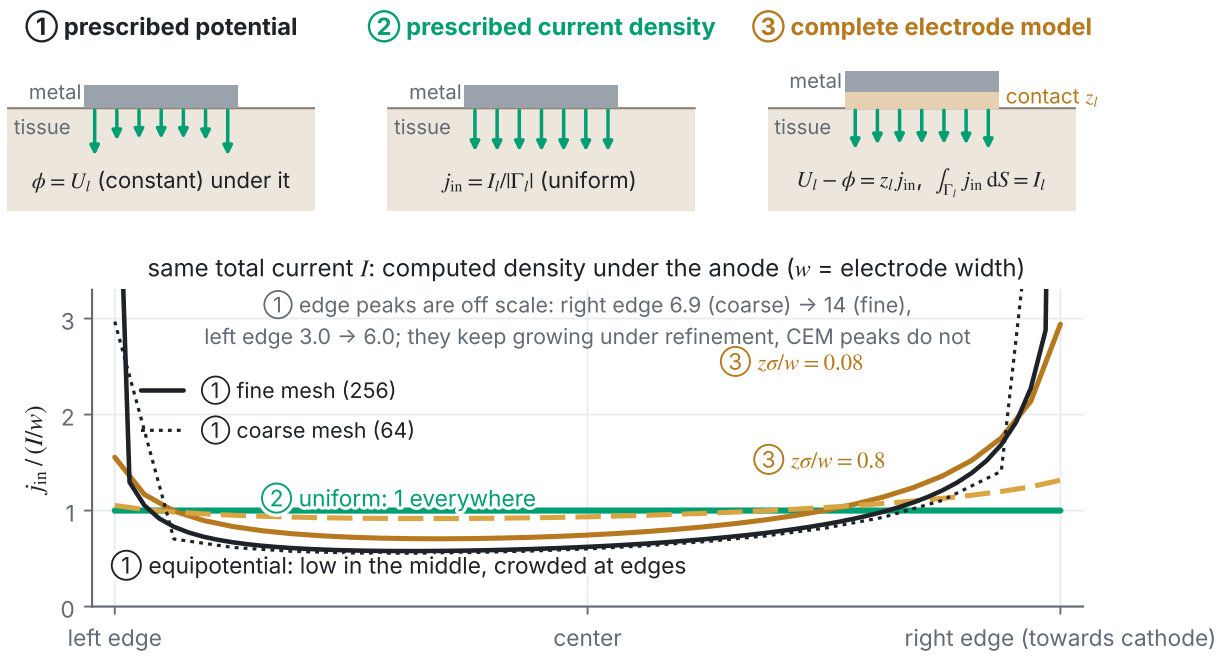

① prescribed potential
② prescribed current density
③ complete electrode model
metal
tissue
$\phi = U_l$ (constant) under it
metal
tissue
$j_{\text{in}} = I_l/|\Gamma_l|$ (uniform)
metal
contact z_l
tissue
$U_l - \phi = z_l j_{\text{in}}, \quad \int_{\Gamma_l} j_{\text{in}}\,\mathrm{d}S = I_l$
same total current I: computed density under the anode (w = electrode width)
① edge peaks are off scale: right edge 6.9 (coarse) → 14 (fine),
left edge 3.0 → 6.0; they keep growing under refinement, CEM peaks do not
③ $z\sigma/w = 0.08$
① fine mesh (256)
① coarse mesh (64)
③ $z\sigma/w = 0.8$
② uniform: 1 everywhere
① equipotential: low in the middle, crowded at edges
$j_{\text{in}} / (I/w)$
3
2
1
0
left edge
center
right edge (towards cathode)